

Static IVM Field Learning

Daniel Ari Friedman

ORCID: 0000-0001-6232-9096

Email: daniel@activeinference.institute

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1 Static IVM Field Learning

1.1 Overview

This section develops machine learning on a static synergetic geometry: scalar fields defined over the Isotropic Vector Matrix (IVM) lattice, learned from sparse noisy observations by Laplacian-regularized kernel-weighted least squares on the lattice graph. All methods live in the `ivm_field.py` module (`quadray.IVMField`, `quadray_shell_norm`, `shell_sites`, `fit_geometry`) and reuse the quadray machinery of `quadray.py` described in [Quadray Methods](#). The learner uses `numpy` only — no machine-learning frameworks — and is fully deterministic for a fixed observation set.

1.2 The IVM Lattice in Quadray Coordinates

Quadray coordinates represent IVM close-packed sphere centers as non-negative integer quadrays normalized so the minimum component is zero (see [Quadray Methods](#)). Two facts organize the lattice. First, normalization adds or subtracts (k, k, k, k) , so the component sum is a projective invariant modulo 4. Second, the IVM sites are exactly one of the four residue classes:

$$q \text{ is an IVM site} \iff \sum_i q_i \equiv 0 \pmod{4}, \quad (1)$$

implemented as `is_ivm_site()` in `ivm_field.py`. The remaining three cosets are the octahedral ($\text{sum} \equiv 2$) and tetrahedral ($\text{sum} \equiv 1, 3$) voids of the packing — lattice points of the ambient grid, but not sphere centers.

The shell norm of a site is the L^1 magnitude of its sum-zero (Coxeter:4D hyperplane) representative. With $s = \sum_i q_i$:

$$N(q) = \sum_{i=1}^4 \left| q_i - \frac{s}{4} \right| = 2k, \quad k \in \mathbb{Z}_{\geq 0}, \quad (2)$$

computed by `quadray_shell_norm()`. Shell k of the lattice is the set of sites with $N(q) = 2k$; `shell_sites(k)` enumerates it by scanning the bounding box $[0, 2k]^4$, filtering on the membership condition (1) and the norm (2), and sorting lexicographically — a deterministic site order. The shell cardinalities are the cuboctahedral numbers:

$$|\{q : N(q) = 2k\}| = 10k^2 + 2, \quad k \geq 1, \quad (3)$$

giving the sequence 1, 12, 42, 92, 162, ... for $k = 0, 1, 2, 3, 4$ — the center plus the cuboctahedral numbers, with shell 1 the twelve-around-one vector equilibrium of **Quadray Methods**. `shell_cardinalities()` counts shells of an enumerated ball, and the test suite pins the sequence exactly.

1.3 The Field Model

`IVMField.lattice_ball(radius)` stores a scalar field over the lattice ball $B_R = \{q : N(q) \leq 2R\}$ as a one-dimensional numpy array keyed by the deterministic site index of `shell_ball_sites()` (shell-major, lexicographic within shell), with a dictionary mapping each normalized site to its array position. Two lattice-graph ingredients drive learning. The adjacency structure uses the twelve IVM neighbor moves — the permutations of $(2, 1, 1, 0)$ collected in `IVM_NEIGHBOR_STEPS` — so two sites in the ball are graph-adjacent exactly when one step of close packing separates them. The graph Laplacian L (method `IVMField._laplacian()`) is the symmetric difference operator on that adjacency:

$$(Lf)_i = \deg(i) f_i - \sum_{j \sim i} f_j. \quad (4)$$

1.4 Learning: Laplacian-Regularized Kernel-Weighted Least Squares

Observations y_j arrive at a sampled subset Ω of sites. Graph distances $d(\cdot, \cdot)$ are hop counts from a multi-source BFS (`_multi_source_distances()` in `ivm_field.py`), and the Gaussian kernel over hops with width τ (`kernel_width`) is:

$$K(d) = \exp(-(d/\tau)^2), \quad d(i, j) = \text{hop distance}. \quad (5)$$

`IVMField.learn()` minimizes a two-regime objective over the ball, with Laplacian regularization weight $\lambda \geq 0$:

$$\min_f \sum_{i \in \Omega} (f_i - y_i)^2 + \sum_{i \notin \Omega} c_i (f_i - t_i)^2 + \lambda \sum_{i \sim j} (f_i - f_j)^2, \quad (6)$$

with three data-fidelity terms: observed sites are pinned to their data with unit weight (no self-smoothing of real observations); unobserved sites carry a kernel confidence weight and a Nadaraya-Watson kernel target,

$$c_i = \max(K(\min_{j \in \Omega} d(i, j)), \varepsilon), \quad t_i = \frac{\sum_{j \in \Omega} K(d(i, j)) y_j}{\sum_{j \in \Omega} K(d(i, j))}, \quad (7)$$

where ε — the `_WEIGHT_FLOOR` constant, 10^{-12} , in `ivm_field.py` — is a numerical floor keeping the normal-equation matrix of (6) positive definite on the connected lattice ball. The normal equations of (6) are

$$(C + \lambda L)f = Ct, \quad (8)$$

with $C = \text{diag}(c_i)$ and t the target vector; `IVMField.learn()` solves them with a dense `numpy.linalg.solve`. The Laplacian term propagates field structure from observed sites across the lattice graph, and the confidence weighting degrades gracefully to pure Laplacian propagation as observation density falls. Because observed rows carry unit weight, the estimator interpolates exact data as $\lambda \rightarrow 0$ and denoises noisy data for moderate λ — both behaviors are verified numerically in `tests/test_ivm_field.py` with fixed seeds.

Two structural facts justify the estimator. A linear field in the embedded XYZ coordinates (the `to_xyz()` image of `quadray.py`) is harmonic on the IVM graph — the twelve neighbor moves sum to zero — so the harmonic extension implied by (8) reproduces it exactly from boundary data (asserted to 10^{-5} in the test suite). And for noisy smooth fields, the learned field tracks the kernel-weighted local mean, which averages observation noise across graph neighborhoods:

$$\text{MSE}(f) = \frac{1}{|B_R|} \sum_{i \in B_R} (f_i - f_i^{\text{truth}})^2, \quad (9)$$

the quantity returned by `IVMField.score()`.

1.5 Recovering Tetrahedral Geometry from Noisy Points

`fit_geometry()` recovers the orientation and scale of a tetrahedron from noisy 3D point clouds, using the quadray basis of `quadray.py`. The four canonical vertex images are $v_i = \text{to_xyz}(e_i)$, where e_i are the unit quadray axes mapped through the 3×4 embedding matrix of `quadray.py`. Given labeled observations $p_m^{(i)} \approx R v_i + o$ (noisy samples of vertex i under rotation R and offset o), vertex centroids c_i are formed and the linear map G is fit in closed form:

$$G = \arg \min_{M \in \mathbb{R}^{3 \times 3}} \sum_{i=1}^4 \|M v_i - c_i\|^2, \quad \text{scale} = \text{sign}(\det G) |\det G|^{1/3}. \quad (10)$$

The least-squares solve uses `numpy.linalg.lstsq` on the transpose system, so noisy multi-sample vertices average out; the residual reported by `TetrahedronFit.residual` is the RMS vertex-centroid mismatch. The test suite recovers a rotated, scaled tetrahedron to 10^{-6} from $\sigma = 10^{-7}$ noise, verifies the signed scale for reflected (negative determinant) fits, and checks that doubling the embedding halves the recovered matrix — orientation and scale separate cleanly.

1.6 Demonstration

The script `quadmath/scripts/ivm_field_demo.py` (run with `MPLBACKEND=Agg`, seed 12) renders the pipeline on the radius-3 ball (147 lattice sites): the synthetic field (a harmonic linear part plus a weak quadratic bowl), the noisy observations at 74 of the 147 sites, and the learned field. The learned reconstruction beats the raw observation noise (reconstruction MSE 0.0558 against observation-noise MSE 0.0720, printed by the demo):

1.7 Cross-References

- Coordinate foundations and the twelve-around-one shell: [Quadray Methods](#)
- Optimization methods on tetrahedral lattices: [Optimization in 4D](#)
- Applications and generalizations: [Extensions](#)

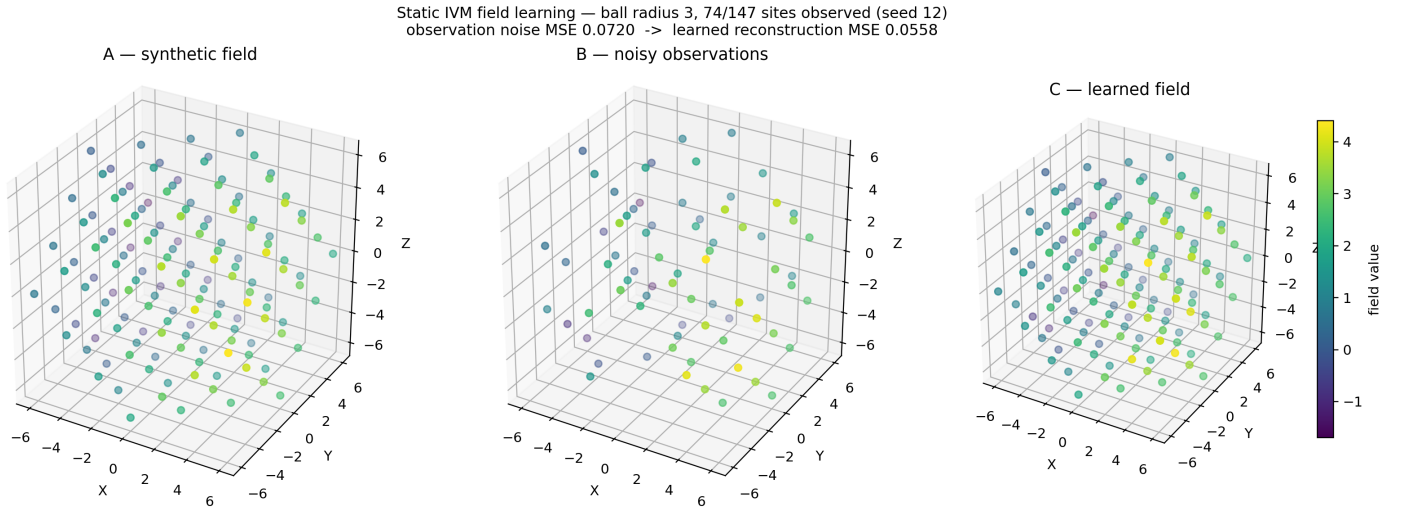


Figure 1: Static IVM field learning on the radius-3 IVM lattice ball (147 sites, shown at their XYZ embedding from the quadray coordinates; axes in embedding units). Panel A: the synthetic ground-truth field — a harmonic linear part plus a weak quadratic bowl — with dot color giving field value on the shared color bar (right). Panel B: the noisy observations at 74 of the 147 sites (seed 12). Panel C: the learned field from Laplacian-regularized kernel-weighted least squares (Eqs. (6)–(8)); reconstruction MSE 0.0558 against observation-noise MSE 0.0720. Reproduce with `quadmath/scripts/ivm_field_demo.py` (MPLBACKEND=Agg, seed 12).